Podcasts and Review of Related Literature

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Today's students operate in a digital and global environment dominated by digital devices such as cell phones, personal digital assistants (PDAs), and mobile devices—and so does today's business (Coleman, 2005). Everything is technologically sophisticated and connected, including computers, cell phones, PDA, iPods, and other MP3 players. These digital devices are capable of receiving e-mail, instant messages, discussion board posts, blogs, and Web conference messages. In recent years, these digital devices have undergone significant advances, enabling podcasting audio/video as well as downloading podcasts from the Internet. This is an area of particular concern for schools, as they must be ready to respond to the cultural shifts of digital technology, and Holley (2006) has also echoed concerns about technology's impact on schools. Many other studies on technology-based learning have also shown that students are likely to focus on the best resources for learning. Aytekin (2005) has stated that students learn best when they actively construct their own understanding in addition to reflective and more responsive forms of education.

The purpose of this study is to determine whether podcasts can enhance student performance (or learning) and how they can be used to do so. More importantly, does the use of podcasts improve students' math and reading achievement? In seeking answers to these questions, it is important for us as educators to review previous studies and related literature on the relationship between podcasts and student performance. If we should, in fact, find a correlation between podcasts (or technology-based education) and student performance, we might then be interested in exploring how podcasts can be used to enhance student performance.

One such study examines instructional technologies that support students' learning preferences. Students prefer to use technology to communicate with each other and with their instructor. Communication such as E-mail, discussion boards, instant messaging, blogs, and conference systems have created more efficient and effective learning and teaching environments. Advances in communication technologies have made podcasting possible through personal digital devices such as iPods and other MP3 players (Ficek, 2006).

In recent years, we have seen that cell phones, PDAs, and the iPod (digital media player) are becoming much more advanced and more than just personal digital devices/media. These digital devices are now capable of receiving digital media such as video/audio podcasts distributed over the Internet for playback. Furthermore, these digital devices and the Internet are very much a part of people's lives.

How do podcasts change the way teachers teach, and students learn? Podcasts allow teachers to easily publish lectures or photos with a myriad of other course content. In addition, podcasts also enable students to listen to or view lectures and lessons on demand. Most students and teachers have access to cell phones and the Internet, and these digital devices are becoming part of people's lives. For this reason, it is in schools' best interest to determine how to integrate technology into the learning environment.

What is a podcast? A podcast is derived from Apple's portable music player, the iPod, and provides a capable delivery of educational content for listening or viewing on students' mobile devices, PDAs, and personal computers. Podcasting is a video/audio file feed in the Really Simple Syndication (RSS) format, a family of web feed formats used to publish frequently updated digital content, such as blogs or podcasts.

What is Podcasting? "Podcasting is a term used to describe a collection of technologies for automatically distributing audio and video programs over the Internet. Podcasting enables independent producers to create self-published, syndicated "radio shows," and gives broadcast radio or television programs a new distribution method. Any digital audio player or computer with audio-playing software can play podcasts. The term "podcast", however, still refers largely to audio content distribution. A podcast is not the same as a webcast, which normally refers to a show distributed by streaming media." from Wikipedia, the free encyclopedia, http://en.wikipedia.org/wiki/Podcast.

Having worked in the technology industry for many years, I can see that technology-based education has dramatically improved during the past decade since its inception. Today's technology-based education can include a personal computer, a PDA, wireless networking, a cell phone, an iPod, or any application software or digital devices. I believe that technology is becoming a part of people's lives. Technology is a new way for people to communicate and interconnect. I can see that there is a new movement in podcasts, a new generation of video/audio delivery morphed into digital technologies. Technology-based education and podcasts via mobile devices can be used for both teaching and learning.

According to Frey and Birnbaum (2002), computer-assisted instruction has a positive effect on lectures, especially in taking notes and studying for exams. In addition, research suggests that a technology-based instructional approach is perceived favorably by students and yields significant improvements in student learning, as evidenced by both student self-reports and objective outcome testing (Smith & Woody, 2000).

The World Wide Web has also created an information-rich teaching and learning environment that facilitates social interaction and exchange among students and instructors.

It allows teachers and students to access a wealth of knowledge-based information and assessment tools that replicate and expand the traditional classroom. It has proven effective for storing, disseminating, and retrieving course-relevant information, and it is available to students anytime, anywhere (Aggarwal & Bento, 2000).

Given this varied result, increasing number of teachers are using technology-assisted instruction in their classes. Many studies have shown that technology-based instruction enhances student performance. However, there is no answer to the question of whether the use of podcasts improves students' math and reading achievement. Part of the reason is that iPods and related MP3 players are still relatively new technology; therefore, further study is needed to assess possible long-term effects. However, there is strong evidence of a clear correlation between students' academic achievement and technology-based learning.

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